EEE407 Tutorial 3c

1. A single-phase circuit runs in single-core 70°C thermoplastic insulated and sheathed cables clipped direct and not bunched with cables of other circuits. If the conductors are copper, the cross-sectional area of the phase conductors is 4 mm^2, the protective conductor is 2.5 mm^2, total length is 45 m, and earth fault loop impedance Z_e is 0.35 ohm, calculate the total loop impedance. Use Table 1, and the multiplier is 1.2 for 70°C thermoplastic insulation, 1.28 for 90°C thermosetting, and 1.04 if the armour of the cable is used as the cpc.

Table 1

mΩ/m at 20°C

Conductor (mm^2)	Resistance (mΩ/m)
1.0	18.1
1.5	12.1
2.5	7.41
4.0	4.61
6.0	3.08
10.0	1.83
16.0	1.15
25.0	0.727
35.0	0.524

2. A single-phase ring circuit is run in 2.5 mm^2 70°C thermoplastic insulated and sheathed cable, while the armoring will provide the function of the cpc at 5.3 mΩ/m. If length is 65 m and Z_e = 0.35 ohm, calculate the total loop impedance. Use Table 1, and the multiplier is 1.2 for 70°C thermoplastic insulation, 1.28 for 90°C thermosetting, and 1.04 if the armour of the cable is used as the cpc. Consider a diversity factor of 0.25 for a ring circuit.

3. A single-phase circuit has I_b = 17 A and is to be wired in flat two-core (with cpc) 70° C thermoplastic insulated and sheathed cable having copper conductors, grouped with four other similar cables, all clipped direct.

   If t_a = 45°C and the circuit is to be protected against both overload and short circuit by a semienclosed fuse to BS 3036, what should be the nominal current rating of that fuse and the minimum cross-sectional area of cable conductor? Use Table 41.2, 4B1, 4C1, and 4D5.

TABLE 41.2 –
Maximum earth fault loop impedance (Z_s) for fuses, for 0.4 s disconnection time with U_0 of 230 V (see Regulation 411.4.201)

(a) General purpose (gG) and motor circuit application (gM) fuses to BS 88-2 – fuse systems E (bolted) and G (clip-in)											
Rating (amperes)	2	4	6	10	16	20	25	32	40	50	63
Z_s (ohms)	33.1	15.6	7.80	4.65	2.43	1.68	1.29	0.99	0.75	0.57	0.44

(b) Fuses to BS 88-3 fuse system C						
Rating (amperes)	5	16	20	32	45	63
Z_s (ohms)	9.93	2.30	1.93	0.91	0.57	0.36

(c) Fuses to BS 3036						
Rating (amperes)	5	15	20	30	45	60
Z_s (ohms)	9.10	2.43	1.68	1.04	0.56	0.40

(d) Fuses to BS 1362		
Rating (amperes)	3	13
Z_s (ohms)	15.6	2.30

TABLE 4B1 – Rating factors (C_a) for ambient air temperatures other than 30 °C

Ambient temperature[a] °C	Insulation				
	60 °C thermosetting	70 °C thermoplastic	90 °C thermosetting	Mineral[a]	
				Thermoplastic covered or bare and exposed to touch 70 °C	Bare and not exposed to touch 105 °C
25	1.04	1.03	1.02	1.07	1.04
30	1.00	1.00	1.00	1.00	1.00
35	0.91	0.94	0.96	0.93	0.96
40	0.82	0.87	0.91	0.85	0.92
45	0.71	0.79	0.87	0.78	0.88
50	0.58	0.71	0.82	0.67	0.84
55	0.41	0.61	0.76	0.57	0.80
60	–	0.50	0.71	0.45	0.75
65	–	–	0.65	–	0.70
70	–	–	0.58	–	0.65
75	–	–	0.50	–	0.60
80	–	–	0.41	–	0.54
85	–	–	–	–	0.47
90	–	–	–	–	0.40
95	–	–	–	–	0.32

a For higher ambient temperatures, consult manufacturer.

▼ **Table 4C1 of BS 7671** Rating factors for one circuit or one multicore cable or for a group of circuits, or a group of mulitcore cables, to be used with current-carrying capacities of Tables 4D1A to 4J4A

Item	Arrangement (cables touching)	Number of circuits or multicore cables												To be used with current-carrying capacities, Reference Method
		1	2	3	4	5	6	7	8	9	12	16	20	
1	Bunched in air, on a surface, embedded or enclosed	1.00	0.80	0.70	0.65	0.60	0.57	0.54	0.52	0.50	0.45	0.41	0.38	A to F
2	Single layer on wall or floor	1.00	0.85	0.79	0.75	0.73	0.72	0.72	0.71	0.70	0.70	0.70	0.70	C
3	Single layer multicore on a perforated horizontal or vertical cable tray system	1.00	0.88	0.82	0.77	0.75	0.73	0.73	0.72	0.72	0.72	0.72	0.72	E
4	Single layer multicore on cable ladder system or cleats etc.	1.00	0.87	0.82	0.80	0.80	0.79	0.79	0.78	0.78	0.78	0.78	0.78	

TABLE 4D5 – 70 °C thermoplastic insulated and sheathed flat cable with protective conductor (COPPER CONDUCTORS)

CONDUCTORS

CURRENT-CARRYING CAPACITY (amperes) and VOLTAGE DROP (per ampere per metre):

Ambient temperature: 30 °C
Conductor operating temperature: 70 °C

Conductor cross-sectional area	Method 100# (above a plasterboard ceiling covered by thermal insulation not exceeding 100 mm in thickness)	Method 101# (above a plasterboard ceiling covered by thermal insulation exceeding 100 mm in thickness)	Method 102# (in a stud wall with thermal insulation with cable touching the inner wall surface)	Method 103# (in a stud wall with thermal insulation with cable not touching the inner wall surface)	Reference Method C* (clipped direct)	Reference Method A* (enclosed in conduit in an insulated wall)	Voltage drop (per ampere per metre)
1	2	3	4	5	6	7	8
(mm^2)	(A)	(A)	(A)	(A)	(A)	(A)	(mV/A/m)
1	13	10.5	13	8	16	11.5	44
1.5	16	13	16	10	20	14.5	29
2.5	21	17	21	13.5	27	20	18
4	27	22	27	18.5	37	26	11
6	34	27	35	23.5	47	32	7.3
10	45	36	47	32	64	44	4.4
16	57	46	63	42.5	85	57	2.8

4. A circuit supplies a heater without pump with a loading of 6 kW at 230 V via cables insulated with 70°C thermoplastic, laid in a group of 2 cables on the wall, and the surrounding temperature is 25 °C. The cable is not under thermal insulation and BS 3036 re-wireable

fuse (C_f = 0.725) is used. Determine the value of the nominal current rating I_n of the protective device and the value of current I_t tabulated in Appendix 4 of BS 7671 (Table 41.4, 4B1, 4C1, and 52.2) for the type of cable, ambient temperature and installation method concerned.

TABLE 41.4 –
Maximum earth fault loop impedance (Z_s) for fuses, for 5 s disconnection time with U_0 of 230 V (see Regulation 411.4.203)

(a) General purpose (gG) and motor circuit application (gM) fuses to BS 88-2 – fuse systems E (bolted) and G (clip in)								
Rating (amperes)	2	4	6	10	16	20	25	32
Z_s (ohms)	44	21	12	6.8	4.0	2.8	2.2	1.7
continued								
Rating (amperes)	40	50	63	80	100	125	160	200
Z_s (ohms)	1.3	0.99	0.78	0.55	0.42	0.32	0.27	0.18
(b) Fuses to BS 88-3 fuse system C								
Rating (amperes)	5	16	20	32	45	63	80	100
Z_s (ohms)	14.6	3.9	3.2	1.6	1.0	0.68	0.51	0.38
(c) Fuses to BS 3036								
Rating (amperes)	5	15	20	30	45	60	100	
Z_s (ohms)	16.8	5.08	3.64	2.51	1.51	1.07	0.51	
(d) Fuses to BS 1362								
Rating (amperes)	3	13						
Z_s (ohms)	22.0	3.64						

TABLE 4B1 – Rating factors (C_a) for ambient air temperatures other than 30 °C

Ambient temperature[a] °C	**Insulation**				
				Mineral[a]	
	60 °C thermosetting	**70 °C thermoplastic**	**90 °C thermosetting**	**Thermoplastic covered or bare and exposed to touch 70 °C**	**Bare and not exposed to touch 105 °C**
25	1.04	1.03	1.02	1.07	1.04
30	1.00	1.00	1.00	1.00	1.00
35	0.91	0.94	0.96	0.93	0.96
40	0.82	0.87	0.91	0.85	0.92
45	0.71	0.79	0.87	0.78	0.88
50	0.58	0.71	0.82	0.67	0.84
55	0.41	0.61	0.76	0.57	0.80
60	–	0.50	0.71	0.45	0.75
65	–	–	0.65	–	0.70
70	–	–	0.58	–	0.65
75	–	–	0.50	–	0.60
80	–	–	0.41	–	0.54
85	–	–	–	–	0.47
90	–	–	–	–	0.40
95	–	–	–	–	0.32

a For higher ambient temperatures, consult manufacturer.

▼ Table 4C1 of BS 7671 Rating factors for one circuit or one multicore cable or for a group of circuits, or a group of mulitcore cables, to be used with current-carrying capacities of Tables 4D1A to 4J4A

Item	Arrangement (cables touching)	Number of circuits or multicore cables												To be used with current-carrying capacities, Reference Method
		1	2	3	4	5	6	7	8	9	12	16	20	
1	Bunched in air, on a surface, embedded or enclosed	1.00	0.80	0.70	0.65	0.60	0.57	0.54	0.52	0.50	0.45	0.41	0.38	A to F
2	Single layer on wall or floor	1.00	0.85	0.79	0.75	0.73	0.72	0.72	0.71	0.70	0.70	0.70	0.70	C
3	Single layer multicore on a perforated horizontal or vertical cable tray system	1.00	0.88	0.82	0.77	0.75	0.73	0.73	0.72	0.72	0.72	0.72	0.72	E
4	Single layer multicore on cable ladder system or cleats etc.	1.00	0.87	0.82	0.80	0.80	0.79	0.79	0.78	0.78	0.78	0.78	0.78	

▼ Table 52.2 of BS 7671 Cable surrounded by thermal insulation

Length in insulation (mm)	Derating factor
50	0.88
100	0.78
200	0.63
400	0.51

5. A single-phase motor circuit is to be wired in insulated and sheathed single-core cables having copper conductors and thermoplastic insulation, 70°C. The cables are to be installed in free air, horizontal, flat spaced by one diameter on cable supports (Reference Method F). If I_b = 135 A, t_a = 50°C and overload and short circuit protection is to be provided by a BS 88–2.2 'gG' fuse, determine is the minimum current rating for that fuse and the minimum cross-sectional area of the cable conductors that can be used. Use Table 41.4, 4B1 and 4D1A.

TABLE 41.4 –
Maximum earth fault loop impedance (Z_s) for fuses, for 5 s disconnection time with U_0 of 230 V
(see Regulation 411.4.203)

(a) General purpose (gG) and motor circuit application (gM) fuses to BS 88-2 – fuse systems E (bolted) and G (clip in)								
Rating (amperes)	2	4	6	10	16	20	25	32
Z_s (ohms)	44	21	12	6.8	4.0	2.8	2.2	1.7
continued								
Rating (amperes)	40	50	63	80	100	125	160	200
Z_s (ohms)	1.3	0.99	0.78	0.55	0.42	0.32	0.27	0.18
(b) Fuses to BS 88-3 fuse system C								
Rating (amperes)	5	16	20	32	45	63	80	100
Z_s (ohms)	14.6	3.9	3.2	1.6	1.0	0.68	0.51	0.38
(c) Fuses to BS 3036								
Rating (amperes)	5	15	20	30	45	60	100	
Z_s (ohms)	16.8	5.08	3.64	2.51	1.51	1.07	0.51	
(d) Fuses to BS 1362								
Rating (amperes)	3	13						
Z_s (ohms)	22.0	3.64						

TABLE 4B1 – Rating factors (C_a) for ambient air temperatures other than 30 °C

Ambient temperature[a] °C	Insulation				
	60 °C thermosetting	70 °C thermoplastic	90 °C thermosetting	Mineral[a]	
				Thermoplastic covered or bare and exposed to touch 70 °C	Bare and not exposed to touch 105 °C
25	1.04	1.03	1.02	1.07	1.04
30	1.00	1.00	1.00	1.00	1.00
35	0.91	0.94	0.96	0.93	0.96
40	0.82	0.87	0.91	0.85	0.92
45	0.71	0.79	0.87	0.78	0.88
50	0.58	0.71	0.82	0.67	0.84
55	0.41	0.61	0.76	0.57	0.80
60	–	0.50	0.71	0.45	0.75
65	–	–	0.65	–	0.70
70	–	–	0.58	–	0.65
75	–	–	0.50	–	0.60
80	–	–	0.41	–	0.54
85	–	–	–	–	0.47
90	–	–	–	–	0.40
95	–	–	–	–	0.32

a For higher ambient temperatures, consult manufacturer.

TABLE 4D1A – Single-core 70 °C thermoplastic insulated cables, non-armoured, with or without sheath (COPPER CONDUCTORS)

CURRENT-CARRYING CAPACITY (amperes): Ambient temperature: 30 °C; Conductor operating temperature: 70 °C

Conductor cross-sectional area	Reference Method A (enclosed in conduit in thermally insulating wall etc.)		Reference Method B (enclosed in conduit on a wall or in trunking etc.)		Reference Method C (clipped direct)		Reference Method F (in free air or on a perforated cable tray horizontal or vertical)				
							Touching			Spaced by one diameter	
	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC flat and touching	3 or 4 cables, three-phase AC flat and touching or trefoil	2 cables, single-phase AC or DC flat	3 cables, three-phase AC flat	3 cables, three-phase AC trefoil	2 cables, single-phase AC or DC or 3 cables three-phase AC flat	
										Horizontal	Vertical
1	2	3	4	5	6	7	8	9	10	11	12
(mm²)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
1	11	10.5	13.5	12	15.5	14	-	-	-	-	-
1.5	14.5	13.5	17.5	15.5	20	18	-	-	-	-	-
2.5	20	18	24	21	27	25	-	-	-	-	-
4	26	24	32	28	37	33	-	-	-	-	-
6	34	31	41	36	47	43	-	-	-	-	-
10	46	42	57	50	65	59	-	-	-	-	-
16	61	56	76	68	87	79	-	-	-	-	-
25	80	73	101	89	114	104	131	114	110	146	130
35	99	89	125	110	141	129	162	143	137	181	162
50	119	108	151	134	182	167	196	174	167	219	197
70	151	136	192	171	234	214	251	225	216	281	254
95	182	164	232	207	284	261	304	275	264	341	311

6. Four industrial single-phase fan assisted process heaters are to be installed adjacent to each other in a factory. Each one is rated at 48 A/230 V. The furthest heater is some 32 m from a distribution board, housing BS 88-2 fuses, located at the intake position. It has been decided to supply the heaters with PVC singles (70°C thermoplastic) in steel trunking (reference

method B), and part of the run will be through an area where the ambient temperature may reach 35°C. The earthing system is TN–S with a measured Ze of 0.3 Ω. There is spare capacity in the distribution board, and the maximum demand will not be exceeded. Determine the minimum size of live conductors and cpc based on the furthest heater. Also, only one common CPC needs to be used. Use Table 41.4, 4B1, 4C1, 4D1A, and 4D1B.

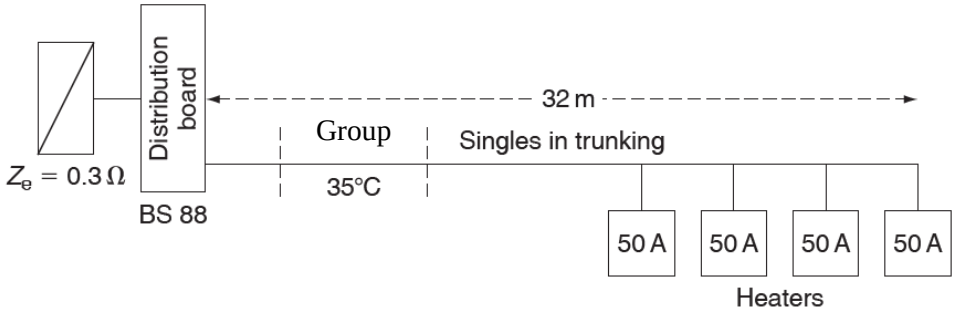


Figure 1

TABLE 41.4 –
Maximum earth fault loop impedance (Z_s) for fuses, for 5 s disconnection time with U_0 of 230 V (see Regulation 411.4.203)

(a) General purpose (gG) and motor circuit application (gM) fuses to BS 88-2 – fuse systems E (bolted) and G (clip in)								
Rating (amperes)	2	4	6	10	16	20	25	32
Z_s (ohms)	44	21	12	6.8	4.0	2.8	2.2	1.7
continued								
Rating (amperes)	40	50	63	80	100	125	160	200
Z_s (ohms)	1.3	0.99	0.78	0.55	0.42	0.32	0.27	0.18

TABLE 4B1 – Rating factors (C_a) for ambient air temperatures other than 30 °C

Ambient temperature[a] °C	Insulation				
				Mineral[a]	
	60 °C thermosetting	70 °C thermoplastic	90 °C thermosetting	Thermoplastic covered or bare and exposed to touch 70 °C	Bare and not exposed to touch 105 °C
25	1.04	1.03	1.02	1.07	1.04
30	1.00	1.00	1.00	1.00	1.00
35	0.91	0.94	0.96	0.93	0.96
40	0.82	0.87	0.91	0.85	0.92
45	0.71	0.79	0.87	0.78	0.88
50	0.58	0.71	0.82	0.67	0.84
55	0.41	0.61	0.76	0.57	0.80
60	–	0.50	0.71	0.45	0.75
65	–	–	0.65	–	0.70
70	–	–	0.58	–	0.65
75	–	–	0.50	–	0.60
80	–	–	0.41	–	0.54
85	–	–	–	–	0.47
90	–	–	–	–	0.40
95	–	–	–	–	0.32

a For higher ambient temperatures, consult manufacturer.

TABLE 4C1 – Rating factors for one circuit or one multicore cable or for a group of circuits, or a group of multicore cables, to be used with current-carrying capacities of Tables 4D1A to 4J4A

Item	Arrangement (cables touching)	Number of circuits or multicore cables												To be used with current-carrying capacities, Reference Method
		1	2	3	4	5	6	7	8	9	12	16	20	
1.	Bunched in air, on a surface, embedded or enclosed	1.00	0.80	0.70	0.65	0.60	0.57	0.54	0.52	0.50	0.45	0.41	0.38	A to F
2.	Single layer on wall or floor	1.00	0.85	0.79	0.75	0.73	0.72	0.72	0.71	0.70	0.70	0.70	0.70	C
3.	Single layer multicore on a perforated horizontal or vertical cable tray system	1.00	0.88	0.82	0.77	0.75	0.73	0.73	0.72	0.72	0.72	0.72	0.72	E
4.	Single layer multicore on cable ladder system or cleats etc.	1.00	0.87	0.82	0.80	0.80	0.79	0.79	0.78	0.78	0.78	0.78	0.78	

TABLE 4D1A – Single-core 70 °C thermoplastic insulated cables, non-armoured, with or without sheath (COPPER CONDUCTORS)

Ambient temperature: 30 °C

CURRENT-CARRYING CAPACITY (amperes): Conductor operating temperature: 70 °C

Conductor cross-sectional area	Reference Method A (enclosed in conduit in thermally insulating wall etc.)		Reference Method B (enclosed in conduit on a wall or in trunking etc.)		Reference Method C (clipped direct)		Reference Method F (in free air or on a perforated cable tray horizontal or vertical)				
							Touching			Spaced by one diameter	
	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC flat and touching	3 or 4 cables, three-phase AC flat and touching or trefoil	2 cables, single-phase AC or DC flat	3 cables, three-phase AC flat	3 cables, three-phase AC trefoil	2 cables, single-phase AC or DC or 3 cables three-phase AC flat	
										Horizontal	Vertical
1	2	3	4	5	6	7	8	9	10	11	12
(mm²)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)	(A)
1	11	10.5	13.5	12	15.5	14	-	-	-	-	-
1.5	14.5	13.5	17.5	15.5	20	18	-	-	-	-	-
2.5	20	18	24	21	27	25	-	-	-	-	-
4	26	24	32	28	37	33	-	-	-	-	-
6	34	31	41	36	47	43	-	-	-	-	-
10	46	42	57	50	65	59	-	-	-	-	-
16	61	56	76	68	87	79	-	-	-	-	-
25	80	73	101	89	114	104	131	114	110	146	130
35	99	89	125	110	141	129	162	143	137	181	162
50	119	108	151	134	182	167	196	174	167	219	197
70	151	136	192	171	234	214	251	225	216	281	254
95	182	164	232	207	284	261	304	275	264	341	311

TABLE 4D1B

VOLTAGE DROP (per ampere per metre): Conductor operating temperature: 70 °C

Conductor cross-sectional area	2 cables, DC	2 cables, single-phase AC									3 or 4 cables, three-phase AC									
		Reference Methods A & B (enclosed in conduit or trunking)			Reference Methods C & F (clipped direct, on tray or in free air)						Reference Methods A & B (enclosed in conduit or trunking)			Reference Methods C & F (clipped direct, on tray or in free air)						
					Cables touching			Cables spaced*						Cables touching, Trefoil			Cables touching, Flat			Cables spaced*, Flat		
1	2	3			4			5			6			7			8			9		
(mm^2)	(mV/A/m)	(mV/A/m)			(mV/A/m)			(mV/A/m)			(mV/A/m)			(mV/A/m)			(mV/A/m)			(mV/A/m)		
1	44	44			44			44			38			38			38			38		
1.5	29	29			29			29			25			25			25			25		
2.5	18	18			18			18			15			15			15			15		
4	11	11			11			11			9.5			9.5			9.5			9.5		
6	7.3	7.3			7.3			7.3			6.4			6.4			6.4			6.4		
10	4.4	4.4			4.4			4.4			3.8			3.8			3.8			3.8		
16	2.8	2.8			2.8			2.8			2.4			2.4			2.4			2.4		
		r	x	z	r	x	z	r	x	z	r	x	z	r	x	z	r	x	z	r	x	z
25	1.75	1.80	0.33	1.80	1.75	0.20	1.75	1.75	0.29	1.80	1.50	0.29	1.55	1.50	0.175	1.50	1.50	0.25	1.55	1.50	0.32	1.55
35	1.25	1.30	0.31	1.30	1.25	0.195	1.25	1.25	0.28	1.30	1.10	0.27	1.10	1.10	0.170	1.10	1.10	0.24	1.10	1.10	0.32	1.15
50	0.93	0.95	0.30	1.00	0.93	0.190	0.95	0.93	0.28	0.97	0.81	0.26	0.85	0.80	0.165	0.82	0.80	0.24	0.84	0.80	0.32	0.86
70	0.63	0.65	0.29	0.72	0.63	0.185	0.66	0.63	0.27	0.69	0.56	0.25	0.61	0.55	0.160	0.57	0.55	0.24	0.60	0.55	0.31	0.63
95	0.46	0.49	0.28	0.56	0.47	0.180	0.50	0.47	0.27	0.54	0.42	0.24	0.48	0.41	0.155	0.43	0.41	0.23	0.47	0.40	0.31	0.51